

Comprehensive Math Refresher & Reference Guide

Audience: Experienced engineer. BS+ background in math through differential equations and linear algebra.

Purpose: Dense reference for solving any standard problem in Algebra, Calculus 1–3, Differential Equations, and Linear Algebra.

Format: Every formula defined. Every technique includes a worked numerical example. Every section includes a Common Mistakes callout.

QUICK LOOKUP: TECHNIQUE SELECTOR

How to use this section: look at the shape of the problem in front of you, match it to a row below, and jump to the cited section for the full method and worked example. This is a diagnostic tool, not the explanation itself — think of it as "if the problem looks like X, go read section Y."

Which Integration Technique?

What you see	Technique	Section
$f(g(x)) \cdot g'(x)$ — composite with its derivative present	u-substitution	3.2
Product of unrelated functions (poly $\times$ trig, poly $\times$ exp, ln, arctan)	Integration by parts (LIATE)	3.2
$\sin^m x \cos^n x, \tan^m x \sec^n x$	Trig integrals	3.2
$\sqrt{a^2 - x^2}, \sqrt{a^2 + x^2}, \sqrt{x^2 - a^2}$	Trig substitution	3.2
Rational function (poly/poly), denominator factorable	Partial fractions (long divide first if $\deg\uparrow \geq \deg\downarrow$)	1.1, 3.2
$e^{ax} \sin(bx)$ or similar cycling product	IBP twice (cycling)	3.2
$\int \ln x$ or $\int \arctan x$ alone	IBP with $dv = dx$	3.2

Decoding the row descriptions:

- " $f(g(x)) \cdot g'(x)$ — composite with its derivative present":** the integrand is one function nested inside another ($f(g(x))$, e.g. $\sin(x^2)$), multiplied by the derivative of that inner function ($g'(x)$, e.g. $2x$). Seeing both the "outside" function and the exact derivative of the "inside" piece is the signal to try *u*-substitution — see the worked example in 3.2.
- "Product of unrelated functions... (LIATE)":** when two factors don't share a chain-rule relationship (e.g. $x^2 \cdot e^x$, not $x \cdot e^{x^2}$), integration by parts applies. **LIATE** is a priority order for picking which factor is *u*: **L**ogarithmic, **I**nverse trig, **A**lgebraic (polynomials), **T**rigonometric, **E**xponential — pick *u* from whichever category comes first in that list, since it should get *simpler* when differentiated.

- " $\sin^m x \cos^n x, \tan^m x \sec^n x$ ": integrands built entirely from powers of trig functions multiplying each other. The exponents m, n tell you which identity to peel off (odd power $\rightarrow$ save one factor and substitute; both even $\rightarrow$ use power-reduction formulas). Full rules in 3.2.
- " $\sqrt{a^2 - x^2}$ " **etc.**: whenever a square root of a sum/difference of squares shows up (and it isn't a perfect-square shortcut), a trig substitution turns the radical into a single trig function via the Pythagorean identities.
- "**deg** $\uparrow \geq$ **deg** $\downarrow$ ": shorthand for "degree of the numerator is greater than or equal to the degree of the denominator" — in that case do polynomial long division *before* setting up partial fractions, otherwise the decomposition breaks.
- "**Cycling product**": IBP that, after being applied twice, produces the *original integral* again on the right-hand side (with a different sign/coefficient) — you then solve for it algebraically instead of integrating further. Classic example: $\int e^x \sin x \, dx$.

Which ODE Method?

An ODE (ordinary differential equation) is "first-order" if it only involves y and y' (no y''), and "second-order" if y'' appears. Match the algebraic shape of your equation to a row.

ODE form	Method	Section
$\frac{dy}{dx} = g(x)h(y)$	Separable	5.1
$y' + P(x)y = Q(x)$	Integrating factor	5.1
$M \, dx + N \, dy = 0$ with $M_y = N_x$	Exact	5.1
$y' + P(x)y = Q(x)y^n$	Bernoulli substitution $v = y^{1-n}$	5.1
$ay'' + by' + cy = 0$	Characteristic equation	5.2
$ay'' + by' + cy = g(x)$, g is poly/exp/trig	Undetermined coefficients	5.2
$ay'' + by' + cy = g(x)$, general g	Variation of parameters	5.2
IVP, need exact solution from initial conditions	Laplace transform	5.3
System $\mathbf{x}' = A\mathbf{x}$	Eigenvalue method	5.4

Decoding the row descriptions:

- " $\frac{dy}{dx} = g(x)h(y)$ " (**separable**): the right side factors into a piece that's purely a function of x times a piece that's purely a function of y — so you can physically separate the variables onto opposite sides of the equation and integrate each side independently.
- " $y' + P(x)y = Q(x)$ " (**linear, integrating factor**): y and y' each appear to the first power only (no y^2 , $y \cdot y'$, etc.), with coefficients that are functions of x . Multiplying by the right "integrating factor" $\mu(x)$ makes the left side collapse into a single derivative $\frac{d}{dx}[\mu y]$.
- " $M \, dx + N \, dy = 0$ with $M_y = N_x$ " (**exact**): rewrite the ODE with all terms on one side as $M(x, y) \, dx + N(x, y) \, dy = 0$. If the partial derivative of M with respect to y equals the partial derivative of N with respect to x , the equation is "exact" and comes from a single potential function $F(x, y) = C$ you can reconstruct directly.
- "**Bernoulli**": looks like the linear case but with a y^n tacked onto the right side, making it nonlinear. The substitution $v = y^{1-n}$ converts it back into a linear equation you can solve with an integrating

factor.

- **"IVP" (initial value problem):** an ODE plus starting conditions like $y(0) = 1$. Laplace transforms are especially convenient here because they bake the initial conditions directly into the algebra instead of requiring you to solve for constants at the end.
- **"System $\mathbf{x}' = A\mathbf{x}$ ":** several first-order ODEs coupled together, written in matrix form. The eigenvalues/eigenvectors of A directly give the solution's building blocks (see 5.4 and its cross-reference to eigenvector mechanics in 6.5).

Which Coordinate System (Calc 3)?

Applies to double/triple integrals over regions — pick the coordinate system that makes the region's boundaries (and thus the limits of integration) as simple as possible.

Region shape	Best coordinates	Section
Rectangular box or general region	Cartesian	4.5
Cylinder, cone, disk in xy -plane	Cylindrical (r, θ, z)	4.5
Sphere, cone from origin	Spherical (ρ, ϕ, θ)	4.5

Decoding the row descriptions: Cylindrical is just polar coordinates (r, θ) in the xy -plane with an ordinary z tacked on — use it whenever the region has circular symmetry around an axis. Spherical adds a second angle ϕ measured down from the z -axis and a single radial distance ρ from the origin — use it whenever the region has symmetry around a *point* (a full or partial ball) rather than an axis.

Which Vector Calculus Theorem?

These four theorems all do the same conceptual trick — trade an integral over a boundary (curve or surface) for an integral over the region it encloses, or vice versa. Which one applies depends on the dimension and whether the boundary is closed.

Situation	Theorem	Section
2D closed curve, want to convert to area integral	Green's	4.6
3D closed curve or want to swap surfaces	Stokes'	4.6
Closed surface, want outward flux	Divergence	4.6
Path-independent field, need work	FT for Line Integrals	4.6

Decoding the row descriptions:

- **Green's:** for a closed loop that lives flat in the xy -plane, converts a line integral around the loop into a double integral over the flat region it bounds.
- **Stokes':** the 3D generalization — converts a line integral around a closed curve (that may bend through space) into a surface integral of the curl over *any* surface whose edge is that curve. Useful for swapping a hard surface for an easier one with the same boundary.
- **Divergence:** for a closed surface (one that fully encloses a solid, like a sphere or box — no edges), converts the outward flux integral into a triple integral of the divergence over the enclosed volume.

- **"Path-independent field, need work":** if a vector field is conservative (curl-free, has a potential function f), the Fundamental Theorem for Line Integrals says the work done only depends on the start/end points, not the path — so you can skip the line integral entirely and just evaluate f at the endpoints.
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SUBJECT 1: ALGEBRA

1.1 Factoring and Algebraic Manipulation

Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Example: $x^2 - 9 = (x + 3)(x - 3)$

Perfect Square Trinomial

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

Example: $x^2 + 6x + 9 = (x + 3)^2$

Sum/Difference of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example: $8x^3 - 27 = (2x - 3)(4x^2 + 6x + 9)$

General Trinomial Factoring

For $ax^2 + bx + c$: find two numbers that multiply to ac and add to b . Split the middle term and factor by grouping.

Example: $6x^2 + 7x - 3$. Need two numbers multiplying to $6 \cdot (-3) = -18$, adding to 7: that's 9 and -2.

$$6x^2 + 9x - 2x - 3 = 3x(2x + 3) - 1(2x + 3) = (3x - 1)(2x + 3)$$

Factoring by Grouping

Example: $x^3 + 2x^2 - 3x - 6 = x^2(x + 2) - 3(x + 2) = (x^2 - 3)(x + 2)$

Completing the Square

To convert $ax^2 + bx + c$:

1. Factor out a from the x terms: $a \left(x^2 + \frac{b}{a}x \right) + c$
2. Add and subtract $\left(\frac{b}{2a} \right)^2$ inside: $a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}$

Example: $2x^2 - 8x + 5$

$$= 2(x^2 - 4x) + 5 = 2(x^2 - 4x + 4 - 4) + 5 = 2(x - 2)^2 - 8 + 5 = 2(x - 2)^2 - 3$$

Use cases: vertex form of parabola, trig substitution setup, integrating $\int \frac{dx}{x^2+bx+c}$.

Polynomial Long Division

Divide $2x^3 - 3x^2 + x - 5$ by $x - 2$:

$$\begin{array}{r}
 2x^2 + x + 3 \\
 x - 2 \overline{) 2x^3 - 3x^2 + x - 5} \\
 \underline{2x^3 - 4x^2} \\
 x^2 + x \\
 \underline{x^2 - 2x} \\
 3x - 5 \\
 \underline{3x - 6} \\
 1
 \end{array}$$

Result: $2x^2 + x + 3 + \frac{1}{x-2}$

Synthetic Division

Only for dividing by $(x - c)$. For $2x^3 - 3x^2 + x - 5 \div (x - 2)$, use $c = 2$:

$$\begin{array}{r|rrrr}
 2 & 2 & -3 & 1 & -5 \\
 & & 4 & 2 & 6 \\
 \hline
 & 2 & 1 & 3 & 1
 \end{array}$$

Same result: $2x^2 + x + 3$ remainder 1.

Partial Fraction Decomposition

Step 1: If $\deg(\text{numerator}) \geq \deg(\text{denominator})$, do polynomial long division first.

Step 2: Factor denominator completely over the reals.

Step 3: Set up the decomposition:

Factor in denominator	Contribution to decomposition
$(ax + b)$	$\frac{A}{ax + b}$
$(ax + b)^2$	$\frac{A}{ax + b} + \frac{B}{(ax + b)^2}$
$(ax + b)^n$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_n}{(ax + b)^n}$
$ax^2 + bx + c$ (irreducible)	$\frac{Ax + B}{ax^2 + bx + c}$

Example (distinct linear): $\frac{3x + 1}{(x - 1)(x + 2)}$

$$\frac{3x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$$

Multiply through: $3x + 1 = A(x + 2) + B(x - 1)$

Set $x = 1$: $4 = 3A \Rightarrow A = \frac{4}{3}$

Set $x = -2$: $-5 = -3B \Rightarrow B = \frac{5}{3}$

Example (irreducible quadratic): $\frac{x^2 + 2x + 3}{(x + 1)(x^2 + 4)}$

$$= \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 4}$$

Multiply through: $x^2 + 2x + 3 = A(x^2 + 4) + (Bx + C)(x + 1)$

Set $x = -1$: $1 - 2 + 3 = A(5) \Rightarrow A = \frac{2}{5}$

Expand and match coefficients for B and C .

Common Mistakes:

- Forgetting to do long division when $\deg(\text{num}) \geq \deg(\text{denom})$
- Setting up $\frac{Ax+B}{(ax+b)^2}$ for a repeated linear factor instead of two separate terms
- Skipping the irreducible quadratic check — $x^2 + 4$ does NOT factor over the reals

1.2 Exponent and Logarithm Laws

Exponent Laws

Law	Formula
Product	$a^m \cdot a^n = a^{m+n}$
Quotient	$a^m / a^n = a^{m-n}$
Power	$(a^m)^n = a^{mn}$
Product base	$(ab)^n = a^n b^n$

Law	Formula
Negative	$a^{-n} = 1/a^n$
Zero	$a^0 = 1 \ (a \neq 0)$
Fractional	$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

Logarithm Laws

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$\log_b(x^r) = r \log_b x$$

$$\log_b x = \frac{\ln x}{\ln b} \quad (\text{change of base})$$

$$\ln(e^x) = x, \quad e^{\ln x} = x$$

$$\log_b(b^x) = x, \quad b^{\log_b x} = x$$

Common Mistakes:

- $\ln(x + y) \neq \ln x + \ln y$ (no product rule for sums)
- $(\ln x)^2 \neq 2 \ln x$ (the power rule requires the argument to be raised, not the log itself)

1.3 Solving Equations

Quadratic: All Methods

Given $ax^2 + bx + c = 0$:

Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant $\Delta = b^2 - 4ac$: if $\Delta > 0$, two real roots; $\Delta = 0$, one repeated real root; $\Delta < 0$, two complex roots.

Example: $3x^2 - 5x - 2 = 0$

$$x = \frac{5 \pm \sqrt{25 + 24}}{6} = \frac{5 \pm 7}{6}$$

$$x = 2 \text{ or } x = -\frac{1}{3}$$

Rational Equations

Multiply through by LCD. Check for extraneous solutions (values making the original denominator zero).

Example: $\frac{1}{x-2} + \frac{3}{x+1} = \frac{4}{x^2-x-2}$

Note $x^2 - x - 2 = (x - 2)(x + 1)$. Multiply by $(x - 2)(x + 1)$:

$$(x + 1) + 3(x - 2) = 4$$

$$x + 1 + 3x - 6 = 4 \Rightarrow 4x = 9 \Rightarrow x = 9/4$$

Check: $x = 9/4$ doesn't make any denominator zero. ✓

Radical Equations

Isolate the radical, square both sides, check for extraneous solutions.

Example: $\sqrt{2x + 3} = x - 1$

Square: $2x + 3 = x^2 - 2x + 1 \Rightarrow x^2 - 4x - 2 = 0$

$$x = \frac{4 \pm \sqrt{16 + 8}}{2} = 2 \pm \sqrt{6}$$

Check both in original: $x = 2 + \sqrt{6}$ works; $x = 2 - \sqrt{6} \approx -0.449$ gives $\sqrt{} \approx 1.55$ but $x - 1 \approx -1.449$ — extraneous.

Absolute Value Equations

$|f(x)| = c$: if $c > 0$, solve $f(x) = c$ and $f(x) = -c$.

$|f(x)| = |g(x)|$: solve $f(x) = g(x)$ and $f(x) = -g(x)$.

Exponential Equations

Isolate the exponential, apply $\ln$ to both sides.

Example: $3^{2x-1} = 5$

$$(2x - 1) \ln 3 = \ln 5$$

$$x = \frac{\ln 5 / \ln 3 + 1}{2} = \frac{\log_3 5 + 1}{2}$$

Logarithmic Equations

Isolate the log, exponentiate. Check domain (argument must be positive).

Example: $\ln(x + 2) + \ln(x - 1) = \ln(4x)$

$$\ln[(x + 2)(x - 1)] = \ln(4x)$$

$$(x + 2)(x - 1) = 4x$$

$$x^2 + x - 2 = 4x \Rightarrow x^2 - 3x - 2 = 0$$

$$x = \frac{3 \pm \sqrt{17}}{2}$$

Discard negative root (check domain).

1.4 Inequalities

Linear Inequalities

Solve like equations; flip the inequality when multiplying/dividing by a negative number.

Quadratic Inequalities

Solve $ax^2 + bx + c > 0$: find roots, test intervals.

Example: $x^2 - 5x + 6 > 0 \Rightarrow (x - 2)(x - 3) > 0$

Roots: $x = 2, 3$. Test intervals $(-\infty, 2)$, $(2, 3)$, $(3, \infty)$:

- $x = 0: (-2)(-3) = 6 > 0 \checkmark$
 - $x = 2.5: (0.5)(-0.5) < 0 \times$
 - $x = 4: (2)(1) = 2 > 0 \checkmark$
- Solution: $(-\infty, 2) \cup (3, \infty)$

Rational Inequalities

Find critical values (zeros of numerator and denominator), test intervals, exclude denominator zeros from solution.

Absolute Value Inequalities

- $|f(x)| < c: -c < f(x) < c$
- $|f(x)| > c: f(x) < -c \text{ or } f(x) > c$

Common Mistakes:

- Not flipping the inequality when multiplying by a negative
- Including denominator zeros in the solution set of a rational inequality

1.5 Functions

Domain Determination

- Rational: exclude where denominator = 0
- Square roots (even): radicand ≥ 0
- Logarithms: argument > 0
- Compositions: apply restrictions sequentially

Example: Domain of $f(x) = \sqrt{\ln(x-1)}$

Need $x-1 > 0$ (for $\ln$), so $x > 1$. Then need $\ln(x-1) \geq 0$, so $x-1 \geq 1$, so $x \geq 2$.

Domain: $[2, \infty)$

Inverse Functions

1. Verify f is one-to-one (horizontal line test, or confirm strictly monotone)
2. Write $y = f(x)$, swap x and y , solve for y
3. The result is $f^{-1}(x)$

Example: $f(x) = \frac{2x-3}{x+1}$. Swap: $x = \frac{2y-3}{y+1}$

$$x(y+1) = 2y-3 \Rightarrow xy+x = 2y-3 \Rightarrow y(x-2) = -3-x$$

$$f^{-1}(x) = \frac{-x-3}{x-2}$$

Transformations

Starting from $y = f(x)$:

Transformation	Effect
$y = f(x) + k$	Shift up k

Transformation	Effect
$y = f(x - h)$	Shift right h
$y = -f(x)$	Reflect over x -axis
$y = f(-x)$	Reflect over y -axis
$y = af(x), a > 1$	Vertical stretch by a
$y = f(bx), b > 1$	Horizontal compress by b

1.6 Trigonometry — Complete Reference

Unit Circle (Key Values)

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\pi/6$	1/2	$\sqrt{3}/2$	$1/\sqrt{3}$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	$\sqrt{3}/2$	1/2	$\sqrt{3}$
$\pi/2$	1	0	undefined
π	0	-1	0
$3\pi/2$	-1	0	undefined
2π	0	1	0

Reciprocal functions: $\csc = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Angle Addition / Subtraction

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Double Angle Formulas

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Power Reduction (used constantly in integration)

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}, \quad \cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

Half Angle Formulas

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Sign depends on the quadrant of $\theta/2$.

Product-to-Sum

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

Inverse Trig Functions

Function	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$(-\infty, \infty)$	$(-\pi/2, \pi/2)$
$\operatorname{arcsec} x$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R} \setminus (-\pi/2, \pi/2)$

Composition identities:

$$\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}, \quad \cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$$

$$\tan(\arcsin x) = \frac{x}{\sqrt{1-x^2}}$$

Draw a reference triangle to derive these.

Solving Trig Equations

Find all solutions in $[0, 2\pi)$, then add $2\pi k$ for general solution.

Example: $2 \sin^2 x - \sin x - 1 = 0$

Factor: $(2 \sin x + 1)(\sin x - 1) = 0$

$$\sin x = -1/2 \Rightarrow x = 7\pi/6, 11\pi/6$$

$$\sin x = 1 \Rightarrow x = \pi/2$$

Common Mistakes:

- Dividing by $\sin x$ or $\cos x$ (you lose solutions where they equal zero — factor instead)
- Forgetting that $\arcsin$ only returns values in $[-\pi/2, \pi/2]$; there's a second solution in $[\pi/2, 3\pi/2]$

1.7 Complex Numbers

Rectangular Form: $z = a + bi$, where $i = \sqrt{-1}$, $i^2 = -1$

- **Addition:** $(a + bi) + (c + di) = (a + c) + (b + d)i$
- **Multiplication:** $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$
- **Division:** $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$
- **Conjugate:** $\bar{z} = a - bi$; note $z\bar{z} = a^2 + b^2 = |z|^2$

Polar Form

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

where $r = |z| = \sqrt{a^2 + b^2}$ (modulus) and $\theta = \arg(z) = \arctan(b/a)$ (argument, careful with quadrant).

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

De Moivre's Theorem

$$(re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos(n\theta) + i \sin(n\theta))$$

Example: $(1 + i)^8$. Write $1 + i = \sqrt{2}e^{i\pi/4}$.

$$(1 + i)^8 = (\sqrt{2})^8 e^{i \cdot 8\pi/4} = 16e^{2\pi i} = 16$$

nth Roots

The n nth roots of $re^{i\theta}$ are:

$$z_k = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \dots, n-1$$

SUBJECT 2: CALCULUS 1

2.1 Limits

Limit Laws

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

- $\lim[f \pm g] = L \pm M$
- $\lim[f \cdot g] = L \cdot M$
- $\lim[f/g] = L/M$ (provided $M \neq 0$)
- $\lim[f^n] = L^n$
- $\lim[\sqrt[n]{f}] = \sqrt[n]{L}$ (provided $L > 0$ for even n)

Formal Definition (ϵ - δ)

$\lim_{x \rightarrow a} f(x) = L$ means: for every $\epsilon > 0$, there exists $\delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$.

Proof example: $\lim_{x \rightarrow 3} (2x - 1) = 5$.

$|f(x) - 5| = |2x - 1 - 5| = 2|x - 3|$. Set $2|x - 3| < \epsilon$, so $\delta = \epsilon/2$.

Techniques for Evaluating Limits

Direct substitution: try $f(a)$ first. If defined and continuous, you're done.

Factoring/canceling (0/0):

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

Rationalization:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{4}$$

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ near a and $\lim g = \lim h = L$, then $\lim f = L$.

Example: $\lim_{x \rightarrow 0} x^2 \sin(1/x)$. Since $-1 \leq \sin(1/x) \leq 1$: $-x^2 \leq x^2 \sin(1/x) \leq x^2$. Both squeeze limits = 0, so the limit = 0.

Limits at Infinity

- $\lim_{x \rightarrow \infty} x^n = \infty$ for $n > 0$; $\lim_{x \rightarrow \infty} x^{-n} = 0$
- For rational functions: divide numerator and denominator by the highest power of x in the denominator.

Example: $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{5x^2 + 1} = \lim_{x \rightarrow \infty} \frac{3 - 2/x}{5 + 1/x^2} = \frac{3}{5}$

Indeterminate Forms and Resolutions

Form	Resolution
0/0	Factor, rationalize, or L'Hôpital
∞/∞	Divide by dominant term or L'Hôpital
$0 \cdot \infty$	Rewrite as $0/(1/\infty)$ or $\infty/(1/0)$, then L'Hôpital
$\infty - \infty$	Combine fractions, rationalize

Form	Resolution
$0^0, 1^\infty, \infty^0$	Take $\ln$, convert to $0/0$ or ∞/∞ , then L'Hôpital

L'Hôpital's Rule: If $\lim f/g$ is $0/0$ or $\pm\infty/\infty$, then $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$ (provided the latter exists).

Example: $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1$

Example (1^∞): $\lim_{x \rightarrow 0^+} x^x$. Let $L = \lim x^x$. Then $\ln L = \lim x \ln x = \lim \frac{\ln x}{1/x} \stackrel{H}{=} \lim \frac{1/x}{-1/x^2} = \lim(-x) = 0$. So $L = e^0 = 1$.

Continuity

f is continuous at a iff:

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Types of discontinuity:

- **Removable:** limit exists but $\neq f(a)$ (or $f(a)$ undefined)
- **Jump:** one-sided limits exist but are unequal
- **Infinite:** limit is $\pm\infty$

Intermediate Value Theorem: If f is continuous on $[a, b]$ and $f(a) < k < f(b)$, then $\exists c \in (a, b)$ with $f(c) = k$.

Common Mistakes:

- Applying L'Hôpital when the form is NOT $0/0$ or ∞/∞
- Forgetting to check that the post-L'Hôpital limit actually exists before concluding
- Assuming continuity at a point just because the function is "nice-looking" there

2.2 Derivatives — All Rules and Techniques

Limit Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example: Find $f'(x)$ for $f(x) = x^2 + 3x$.

$$\frac{(x+h)^2 + 3(x+h) - x^2 - 3x}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \rightarrow 2x + 3$$

Differentiation Rules

Rule	Formula	Example
Power	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[x^5] = 5x^4$
Constant multiple	$\frac{d}{dx}[cf] = cf'$	$\frac{d}{dx}[3x^4] = 12x^3$

Rule	Formula	Example
Sum/Difference	$(f \pm g)' = f' \pm g'$	
Product	$(fg)' = f'g + fg'$	$(x^2 \sin x)' = 2x \sin x + x^2 \cos x$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	
Chain	$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$	$\frac{d}{dx}[\sin(x^2)] = \cos(x^2) \cdot 2x$

Triple chain rule example: $\frac{d}{dx}[e^{\sin(x^2)}] = e^{\sin(x^2)} \cdot \cos(x^2) \cdot 2x$

Derivatives of Standard Functions

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arccos x$	$-1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$
$\operatorname{arcsec} x$	$1/(\sqrt{x^2-1})$

Implicit Differentiation

Differentiate both sides with respect to x ; whenever you differentiate a y term, multiply by dy/dx .

Example: Find dy/dx for $x^3 + y^3 = 6xy$.

$$\begin{aligned}
 3x^2 + 3y^2 \frac{dy}{dx} &= 6y + 6x \frac{dy}{dx} \\
 \frac{dy}{dx}(3y^2 - 6x) &= 6y - 3x^2 \\
 \frac{dy}{dx} &= \frac{6y-3x^2}{3y^2-6x} = \frac{2y-x^2}{y^2-2x}
 \end{aligned}$$

Logarithmic Differentiation

Useful when: the function is a product/quotient of many terms, or has variable exponents.

Example: $y = \frac{x^3 \sqrt{x+1}}{(2x-1)^4}$

Take $\ln$: $\ln y = 3 \ln x + \frac{1}{2} \ln(x+1) - 4 \ln(2x-1)$

Differentiate: $\frac{y'}{y} = \frac{3}{x} + \frac{1}{2(x+1)} - \frac{8}{2x-1}$

Multiply both sides by y .

Example (variable exponent): $y = x^x$

$\ln y = x \ln x \Rightarrow \frac{y'}{y} = \ln x + 1 \Rightarrow y' = x^x (\ln x + 1)$

Higher-Order Derivatives

$f'', f''', f^{(n)}$ or $\frac{d^2 y}{dx^2}$, etc. — differentiate repeatedly.

Linear Approximation

$$L(x) = f(a) + f'(a)(x - a)$$

Example: Approximate $\sqrt{4.02}$. Let $f(x) = \sqrt{x}$, $a = 4$:

$$f'(x) = 1/(2\sqrt{x}), f'(4) = 1/4$$

$$L(4.02) = 2 + \frac{1}{4}(0.02) = 2.005$$

Common Mistakes:

- Product rule: writing $(fg)' = f'g'$ instead of $f'g + fg'$
- Chain rule: forgetting to multiply by the derivative of the inner function
- Implicit differentiation: not applying chain rule to y terms (missing the dy/dx factor)

2.3 Applications of Derivatives

Increasing/Decreasing and First Derivative Test

- $f' > 0$ on $(a, b) \rightarrow f$ is increasing on (a, b)
- $f' < 0$ on $(a, b) \rightarrow f$ is decreasing on (a, b)
- At a critical point c where $f'(c) = 0$ or $f'(c)$ DNE:
 - f' changes $+$ $\rightarrow$ $-$: local maximum
 - f' changes $-$ $\rightarrow$ $+$: local minimum
 - No sign change: neither

Concavity and Second Derivative Test

- $f'' > 0$: concave up
- $f'' < 0$: concave down
- $f''(c) = 0$ and sign changes: inflection point
- At a critical point c : $f''(c) > 0 \rightarrow$ local min; $f''(c) < 0 \rightarrow$ local max; $f''(c) = 0 \rightarrow$ inconclusive (use FDT)

Absolute Extrema on $[a, b]$ — Closed Interval Method

1. Find all critical points in (a, b)
2. Evaluate f at each critical point AND at a and b
3. Largest value = absolute max; smallest = absolute min

Example: $f(x) = x^3 - 3x^2 + 1$ on $[-1/2, 4]$

$f'(x) = 3x^2 - 6x = 3x(x - 2)$; critical points $x = 0, 2$

$f(-1/2) = -1/8 - 3/4 + 1 = 1/8$

$f(0) = 1, f(2) = 8 - 12 + 1 = -3, f(4) = 64 - 48 + 1 = 17$

Absolute max: 17 at $x = 4$. Absolute min: -3 at $x = 2$.

Applied Optimization — Setup Procedure

1. Assign variables, draw a diagram
2. Write the objective function (what you're maximizing/minimizing)
3. Write the constraint equation; use it to eliminate one variable
4. Differentiate, set $= 0$, solve
5. Verify it's a max/min (FDT, SDT, or endpoints)
6. Answer the original question (include units)

Example: A box with square base and open top, volume = 32 m^3 . Minimize surface area.

Let base side = x , height = h .

Surface area: $A = x^2 + 4xh$. Constraint: $x^2h = 32 \Rightarrow h = 32/x^2$.

$A(x) = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}$

$A'(x) = 2x - \frac{128}{x^2} = 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$

$h = 32/16 = 2$. Dimensions: $4 \times 4 \times 2 \text{ m}$.

Related Rates — Setup Procedure

1. Draw and label all quantities (with values and rates given)
2. Write an equation relating the quantities
3. Differentiate both sides with respect to t
4. Substitute known values and solve for the unknown rate

Example: Ladder: 10 m ladder, foot sliding away at 2 m/s. How fast is the top sliding down when foot is 6 m from wall?

$x^2 + y^2 = 100$. Differentiate: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$.

At $x = 6$: $y = 8$. Given $dx/dt = 2$.

$2(6)(2) + 2(8) \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{3}{2} \text{ m/s}$ (negative = sliding down).

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Rolle's Theorem: Special case where $f(a) = f(b)$, guaranteeing $f'(c) = 0$.

Common Mistakes:

- Optimization: not checking whether the critical point is a max or min (just assuming)
- Related rates: substituting numerical values BEFORE differentiating (rates must come from the differentiated equation)
- Closed interval method: forgetting to evaluate at endpoints

SUBJECT 3: CALCULUS 2

3.1 Integration — Foundations

Antiderivatives and Indefinite Integrals

$$\int f(x) dx = F(x) + C \text{ where } F'(x) = f(x)$$

Fundamental Theorem of Calculus

Part 1: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

With chain rule: $\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) \cdot g'(x)$

Part 2: $\int_a^b f(x) dx = F(b) - F(a)$

Standard Integral Table

$f(x)$	$\int f(x) dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$1/x$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\tan x$	$\ln \sec x + C$
$\cot x$	$\ln \sin x + C$
$\sec x$	$\ln \sec x + \tan x + C$
$\csc x$	$-\ln \csc x + \cot x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$

$f(x)$	$\int f(x) dx$
$\csc x \cot x$	$-\csc x + C$
$1/\sqrt{1-x^2}$	$\arcsin x + C$
$1/(1+x^2)$	$\arctan x + C$
$1/(x\sqrt{x^2-1})$	$\operatorname{arcsec} x + C$

3.2 Integration Techniques

u-Substitution

When to use: Composite function where $u = g(x)$ simplifies to a known form. Spot the pattern $f(g(x)) \cdot g'(x)$.

Procedure:

1. Choose $u = g(x)$
2. Compute $du = g'(x) dx$, solve for dx
3. Substitute everything — the result should be entirely in terms of u
4. Integrate in u
5. Back-substitute to x

Definite integrals: change limits: new limits are $g(a)$ and $g(b)$.

Example: $\int_0^1 2xe^{x^2} dx$

$u = x^2$, $du = 2x dx$; limits: $u(0) = 0$, $u(1) = 1$

$$\int_0^1 e^u du = e^1 - e^0 = e - 1$$

Example (tricky): $\int \frac{x}{x^2+1} dx$

$u = x^2 + 1$, $du = 2x dx$

$$\int \frac{1}{2u} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln(x^2 + 1) + C$$

Common Mistakes:

- Forgetting to change the limits for definite integrals
- Not fully substituting: leaving an x in the integrand
- Choosing u for the outer function instead of the inner

Integration by Parts

$$\int u dv = uv - \int v du$$

LIATE priority for u : Logarithm, Inverse trig, Algebraic, Trigonometric, Exponential

Example: $\int xe^x dx$

$u = x$, $dv = e^x dx \rightarrow du = dx$, $v = e^x$

$$= xe^x - \int e^x dx = xe^x - e^x + C = e^x(x - 1) + C$$

Example (repeated IBP): $\int x^2 \sin x \, dx$

Apply twice; tabular method:

Sign	u (differentiate)	dv (integrate)
+	x^2	$\sin x$
−	$2x$	$-\cos x$
+	2	$-\sin x$
−	0	$\cos x$

Result: $-x^2 \cos x + 2x \sin x + 2 \cos x + C$

Cycling IBP: $\int e^x \sin x \, dx$

$$u = e^x, dv = \sin x: = -e^x \cos x + \int e^x \cos x \, dx$$

$$\text{Apply again:} = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

$$\text{Let } I = \int e^x \sin x \, dx: I = -e^x \cos x + e^x \sin x - I$$

$$2I = e^x(\sin x - \cos x) \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$$

$$\int \ln x \, dx: u = \ln x, dv = dx: = x \ln x - x + C$$

Trigonometric Integrals

$$\int \sin^m x \cos^n x \, dx:$$

- If m (power of $\sin$) is odd: save one $\sin x$, convert rest using $\sin^2 x = 1 - \cos^2 x$, let $u = \cos x$
- If n (power of $\cos$) is odd: save one $\cos x$, convert rest using $\cos^2 x = 1 - \sin^2 x$, let $u = \sin x$
- If both even: use power reduction identities $\sin^2 = (1 - \cos 2x)/2$, $\cos^2 = (1 + \cos 2x)/2$

Example: $\int \sin^3 x \cos^2 x \, dx$

$$= \int \sin^2 x \cos^2 x \sin x \, dx = \int (1 - \cos^2 x) \cos^2 x \sin x \, dx$$

$$u = \cos x, du = -\sin x \, dx:$$

$$= -\int (1 - u^2)u^2 \, du = -\int (u^2 - u^4) \, du = -\frac{u^3}{3} + \frac{u^5}{5} + C = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

$$\int \tan^m x \sec^n x \, dx:$$

- n even: save $\sec^2 x$, convert rest using $\sec^2 = 1 + \tan^2$, let $u = \tan x$
- m odd: save $\sec x \tan x$, convert rest using $\tan^2 = \sec^2 - 1$, let $u = \sec x$
- m even, n odd: use $\tan^2 = \sec^2 - 1$ to reduce to powers of $\sec x$ alone; integrate via reduction formula or IBP
- $\int \tan^m x \, dx$ alone ($n = 0$): peel off $\tan^2 x = \sec^2 x - 1$ repeatedly. E.g., $\int \tan^3 x \, dx = \int \tan x (\sec^2 x - 1) \, dx = \frac{\tan^2 x}{2} - \ln |\sec x| + C$

Trigonometric Substitution

Integrand form	Substitution	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$

Integrand form	Substitution	Identity used
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Procedure: Substitute, simplify the radical (take positive square root), integrate in θ , back-substitute using a reference triangle.

Example: $\int \frac{\sqrt{9-x^2}}{x^2} dx$

$$x = 3 \sin \theta, dx = 3 \cos \theta d\theta, \sqrt{9-x^2} = 3 \cos \theta$$

$$= \int \frac{3 \cos \theta}{9 \sin^2 \theta} \cdot 3 \cos \theta d\theta = \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta = \int (\csc^2 \theta - 1) d\theta$$

$$= -\cot \theta - \theta + C$$

$$\text{Back-sub: } \sin \theta = x/3, \text{ so } \cos \theta = \sqrt{9-x^2}/3, \cot \theta = \sqrt{9-x^2}/x, \theta = \arcsin(x/3)$$

$$= -\frac{\sqrt{9-x^2}}{x} - \arcsin \frac{x}{3} + C$$

Completing the square first: $\int \frac{dx}{\sqrt{x^2+4x+5}} = \int \frac{dx}{\sqrt{(x+2)^2+1}}$, then $u = x+2$, trig sub $u = \tan \theta$.

Alternative for $\sqrt{x^2 \pm a^2}$: hyperbolic substitution $x = a \sinh u$ often gives a cleaner result — see §3.4b.

Common Mistakes:

- Forgetting dx in terms of $d\theta$ after substitution
- Taking the wrong sign of the square root ($\sqrt{a^2 \cos^2 \theta} = a|\cos \theta|$, assume $\cos \theta > 0$ for $\theta \in (-\pi/2, \pi/2)$)
- Forgetting to back-substitute at the end

3.3 Improper Integrals

Type I (infinite limits):

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

Type II (discontinuous integrand at $c \in [a, b]$):

$$\int_a^b f(x) dx = \lim_{t \rightarrow c^-} \int_a^t f(x) dx + \lim_{t \rightarrow c^+} \int_t^b f(x) dx$$

p-integral results:

$$\int_1^\infty \frac{1}{x^p} dx \text{ converges iff } p > 1, \text{ value} = \frac{1}{p-1}$$

$$\int_0^1 \frac{1}{x^p} dx \text{ converges iff } p < 1$$

Example: $\int_1^\infty \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{x}\right]_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1\right) = 1$

Comparison Test: If $0 \leq f(x) \leq g(x)$ for $x \geq a$:

- $\int g$ converges $\Rightarrow \int f$ converges
- $\int f$ diverges $\Rightarrow \int g$ diverges

Common Mistakes:

- Evaluating $\int_{-1}^1 \frac{1}{x} dx$ directly without splitting at $x = 0$ (the integrand is discontinuous there — the integral diverges)

3.4 Applications of Integration

Area Between Curves

$$A = \int_a^b [f(x) - g(x)] dx \quad \text{where } f(x) \geq g(x) \text{ on } [a, b]$$

If curves cross: split at intersection points.

Volumes — Disk/Washer Method (rotation about horizontal axis)

$$V = \pi \int_a^b [R(x)]^2 dx \quad (\text{disk})$$

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx \quad (\text{washer, outer radius } R, \text{ inner radius } r)$$

Volumes — Shell Method (rotation about vertical axis)

$$V = 2\pi \int_a^b x \cdot f(x) dx$$

When to use: Shell is often easier when washers would require solving for x in terms of y .

Arc Length

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Surface Area of Revolution (about x -axis)

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

Work

$$W = \int_a^b F(x) dx \quad (\text{force varying with position})$$

Common Mistakes:

- Washer method: squaring $(R - r)$ instead of $R^2 - r^2$
- Shell method: forgetting the factor of 2π and the x factor

3.4b Hyperbolic Functions

These appear in Laplace tables, trig substitution alternatives, and solutions to $y'' - k^2 y = 0$. Cross-reference: Laplace pairs in §5.3.

Definitions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x}$$

Key Identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\sinh(2x) = 2 \sinh x \cosh x, \quad \cosh(2x) = \cosh^2 x + \sinh^2 x$$

Derivatives and Integrals

$f(x)$	$f'(x)$	$\int f(x) dx$
$\sinh x$	$\cosh x$	$\cosh x + C$
$\cosh x$	$\sinh x$	$\sinh x + C$
$\tanh x$	$\operatorname{sech}^2 x$	$\ln(\cosh x) + C$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$	$\arctan(\sinh x) + C$

Inverse Hyperbolic (useful integral forms)

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 - a^2}\right) + C \quad (x > a)$$

Connection to trig sub: $\int \frac{dx}{\sqrt{x^2 + a^2}}$ can be done via $x = a \tan \theta$ (giving $\ln |\sec \theta + \tan \theta|$) or via $x = a \sinh u$ — both yield the same result in different form.

3.4c Numerical Integration

Used when an antiderivative doesn't exist in closed form (e.g., $\int e^{-x^2} dx$) or when f is given as data.

Trapezoidal Rule

Divide $[a, b]$ into n equal subintervals of width $h = (b - a)/n$, with nodes $x_0, x_1, \dots, x_n$:

$$\int_a^b f(x) dx \approx \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)]$$

Error bound: $|E_T| \leq \frac{(b-a)^3}{12n^2} \max_{[a,b]} |f''(x)|$. Error is $O(h^2)$.

Simpson's Rule (requires n even)

$$\int_a^b f(x) dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)]$$

Pattern of coefficients: 1, 4, 2, 4, 2, ..., 4, 1

Error bound: $|E_S| \leq \frac{(b-a)^5}{180n^4} \max_{[a,b]} |f^{(4)}(x)|$. Error is $O(h^4)$ — much better than Trapezoidal.

Example: Approximate $\int_0^1 e^{x^2} dx$ with $n = 4$ (Simpson's):

$h = 0.25$, nodes: $x_0 = 0, x_1 = 0.25, x_2 = 0.5, x_3 = 0.75, x_4 = 1$

f values: 1, $e^{0.0625}$, $e^{0.25}$, $e^{0.5625}$, e

$\approx \frac{0.25}{3} [1 + 4(1.0645) + 2(1.2840) + 4(1.7551) + 2.7183] \approx 1.4637$

Common Mistakes:

- Simpson's: using odd n (requires even number of subintervals)
- Trapezoidal: doubling the endpoints (they get coefficient 1, not 2)
- Confusing n (number of subintervals) with number of nodes ($n + 1$)

3.5 Sequences and Series

Sequences

A sequence $\{a_n\}$ converges if $\lim_{n \rightarrow \infty} a_n = L$ (finite). Otherwise it diverges.

Apply L'Hôpital by treating n as continuous: $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$.

Geometric Series

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad \text{iff } |r| < 1$$

Diverges if $|r| \geq 1$.

Convergence Tests

Test	Statement	Best For
Divergence Test	If $\lim a_n \neq 0$, diverges	Quick check; cannot confirm convergence
Integral Test	$\sum a_n$ and $\int f(x) dx$ both converge or both diverge (f positive, decreasing, $a_n = f(n)$)	$a_n = 1/n^p$, etc.
p-Series	$\sum 1/n^p$ converges iff $p > 1$	
Direct Comparison	$0 \leq a_n \leq b_n$: if $\sum b_n$ converges, so does $\sum a_n$	Known comparison series
Limit Comparison	If $\lim a_n/b_n = c > 0$, both series behave the same	Rational-like terms

Test	Statement	Best For
Alternating Series Test	$\sum (-1)^n b_n$ converges if $b_n \geq 0$, decreasing, $b_n \rightarrow 0$	Alternating series
Ratio Test	$L = \lim$	a_{n+1}/a_n
Root Test	$L = \lim$	a_n

Absolute vs. conditional convergence:

- Absolutely convergent: $\sum |a_n|$ converges (implies convergence)
- Conditionally convergent: $\sum a_n$ converges but $\sum |a_n|$ diverges (e.g., alternating harmonic series)

Power Series

$$\sum_{n=0}^{\infty} c_n (x - a)^n$$

Radius of convergence via ratio test: $\frac{1}{R} = \lim \left| \frac{c_{n+1}}{c_n} \right|$

After finding R : check each endpoint $x = a \pm R$ separately (ratio test is inconclusive there).

Taylor and Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Maclaurin series to memorize:

Function	Series	Converges for
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	all x
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	all x
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	all x
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \leq 1$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$ x < 1$
$\arctan x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	$ x \leq 1$

Operations: differentiate or integrate term by term within the radius of convergence. Substitute directly (e.g., e^{-x^2} from e^x).

3.6 Parametric and Polar

Parametric Curves

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{d(dy/dx)/dt}{dx/dt}$$

$$\text{Arc length: } L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Polar Coordinates

$$x = r \cos \theta, y = r \sin \theta, r^2 = x^2 + y^2, \tan \theta = y/x$$

Common curves:

- Circle: $r = a$ (radius a) or $r = 2a \cos \theta$ (center on x -axis)
- Cardioid: $r = a(1 + \cos \theta)$
- Rose: $r = a \cos(n\theta)$ — n petals if n odd, $2n$ if n even
- Lemniscate: $r^2 = a^2 \cos(2\theta)$

Area in polar:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

$$\text{For area between curves: } A = \frac{1}{2} \int_{\alpha}^{\beta} (r_{\text{outer}}^2 - r_{\text{inner}}^2) d\theta$$

Arc length in polar:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Common Mistakes:

- Series: applying the ratio test to get $L = 1$ and concluding convergence (inconclusive!)
- Polar area: using $\int r d\theta$ instead of $\frac{1}{2} \int r^2 d\theta$

SUBJECT 4: CALCULUS 3

4.1 Vectors and 3D Geometry

Vector Operations

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$:

$$\text{Dot product: } \mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

Projection of $\mathbf{b}$ onto $\mathbf{a}$:

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}, \quad \text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \mathbf{a}$$

Cross product:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta = \text{area of parallelogram}$
- $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ (right-hand rule)
- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (anti-commutative)

Lines in 3D

Through point $P_0 = (x_0, y_0, z_0)$ with direction $\mathbf{d} = \langle a, b, c \rangle$:

Parametric: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$

Symmetric: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

Planes

Through P_0 with normal $\mathbf{n} = \langle a, b, c \rangle$:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Distance from point (x_1, y_1, z_1) to plane $ax + by + cz = d$:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Quadric Surfaces (Standard Forms)

Surface	Equation
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Hyperbolic paraboloid (saddle)	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
Cone	$z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Hyperboloid of one sheet	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Common Mistakes:

- Cross product: computing $\mathbf{a} \times \mathbf{b}$ when you need $\mathbf{b} \times \mathbf{a}$ (sign error in normal direction)
- Dot product: getting the angle wrong because $\cos^{-1}$ gives $[0, \pi]$ only

4.2 Vector-Valued Functions

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle, \quad \int \mathbf{r}(t) dt = \left\langle \int f dt, \int g dt, \int h dt \right\rangle + \mathbf{C}$$

Arc length: $L = \int_a^b |\mathbf{r}'(t)| dt$

Unit tangent: $\mathbf{T} = \frac{\mathbf{r}'}{|\mathbf{r}'|}$

Curvature: $\kappa = \frac{|\mathbf{T}'|}{|\mathbf{r}'|} = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$

Principal normal: $\mathbf{N} = \frac{\mathbf{T}'}{|\mathbf{T}'|}$, **Binormal:** $\mathbf{B} = \mathbf{T} \times \mathbf{N}$

Acceleration components:

$$a_T = \frac{d|\mathbf{v}|}{dt} = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|}, \quad a_N = \kappa |\mathbf{v}|^2 = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|}$$

4.3 Partial Derivatives

$f_x = \frac{\partial f}{\partial x}$: differentiate w.r.t. x , treating y as constant

Clairaut's theorem: If f_{xy} and f_{yx} are both continuous, then $f_{xy} = f_{yx}$.

Chain Rule (Multivariable)

If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

If $z = f(x, y)$ and $x = g(s, t)$, $y = h(s, t)$:

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}$$

Implicit differentiation: For $F(x, y) = 0$:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

For $F(x, y, z) = 0$: $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$

Gradient

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

- Points in direction of steepest increase; magnitude = rate of steepest ascent
- Perpendicular to level curves (2D) and level surfaces (3D)

Directional derivative: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ ($\mathbf{u}$ must be a unit vector)

Tangent plane to $z = f(x, y)$ at (x_0, y_0, z_0) :

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

4.4 Multivariable Optimization

Critical points: Solve $\nabla f = \mathbf{0}$, i.e., $f_x = 0$ and $f_y = 0$ simultaneously.

Second derivative test: Compute $D = f_{xx}f_{yy} - (f_{xy})^2$ at each critical point.

D	f_{xx}	Conclusion
$D > 0$	> 0	Local minimum
$D > 0$	< 0	Local maximum
$D < 0$	—	Saddle point
$D = 0$	—	Inconclusive

Lagrange Multipliers

Optimize $f(x, y, z)$ subject to $g(x, y, z) = k$:

$$\nabla f = \lambda \nabla g, \quad g = k$$

This gives a system of equations; solve for x, y, z, λ .

Two constraints: $\nabla f = \lambda \nabla g + \mu \nabla h$

Example: Maximize $f(x, y) = xy$ subject to $x + y = 10$.

$$\nabla f = \langle y, x \rangle = \lambda \langle 1, 1 \rangle = \lambda \nabla(x + y)$$

$y = \lambda, x = \lambda$, so $x = y$. Combined with $x + y = 10$: $x = y = 5$. Max = 25.

Common Mistakes:

- Forgetting to check the boundary when finding absolute extrema on a closed region
- SDT: computing $D = f_{xx}f_{yy} - f_{xy}$ instead of $f_{xx}f_{yy} - (f_{xy})^2$

4.5 Multiple Integrals

Double Integrals — Setup

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Switching order: Sketch the region R , re-describe its bounds with the other variable as the outer integral.

$$\text{Polar: } \iint_R f(x, y) dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

Note: $dA = r dr d\theta$ (don't forget the Jacobian r !)

Triple Integrals

Cylindrical coordinates: $x = r \cos \theta, y = r \sin \theta, z = z; dV = r dr d\theta dz$

Spherical coordinates: $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi; dV = \rho^2 \sin \phi d\rho d\phi d\theta$

- ρ : distance from origin
- ϕ : angle from positive z -axis (polar angle), $\phi \in [0, \pi]$
- θ : azimuthal angle from x -axis, $\theta \in [0, 2\pi]$

When to use: Cylindrical for cylinders/cones; spherical for spheres and cones from origin.

Change of Variables (Jacobian)

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

4.5b Surface Integrals

Required setup knowledge for Stokes' and Divergence theorems (§4.6).

Scalar Surface Integral

For surface S parametrized by $\mathbf{r}(u, v)$ over region D :

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

The cross product $\mathbf{r}_u \times \mathbf{r}_v$ gives the normal vector; its magnitude is the area element.

Vector Surface Integral (Flux)

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA$$

Choose orientation: $\mathbf{r}_u \times \mathbf{r}_v$ points outward (or upward) — flip sign if needed.

Surface $z = g(x, y)$ (most common case)

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot \langle -g_x, -g_y, 1 \rangle dA \quad (\text{upward normal})$$

The area element: $dS = \sqrt{1 + g_x^2 + g_y^2} dA$

Example (flux): Find the upward flux of $\mathbf{F} = \langle x, y, z \rangle$ through $z = 4 - x^2 - y^2$ over $z \geq 0$.

$g_x = -2x$, $g_y = -2y$. Normal vector (upward): $\langle 2x, 2y, 1 \rangle$

$$\mathbf{F} \cdot \langle 2x, 2y, 1 \rangle = 2x^2 + 2y^2 + z = 2x^2 + 2y^2 + (4 - x^2 - y^2) = x^2 + y^2 + 4$$

Convert to polar ($z \geq 0 \Rightarrow r \leq 2$):

$$\int_0^{2\pi} \int_0^2 (r^2 + 4)r dr d\theta = 2\pi \left[\frac{r^4}{4} + 2r^2 \right]_0^2 = 2\pi(4 + 8) = 24\pi$$

Common Parametrizations

Surface	Parametrization	Normal magnitude
Sphere $\rho = a$	$\mathbf{r} = a\langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle$	$a^2 \sin \phi$
Cylinder $r = a, 0 \leq z \leq h$	$\mathbf{r} = \langle a \cos \theta, a \sin \theta, z \rangle$	a
$z = g(x, y)$	$\mathbf{r} = \langle x, y, g(x, y) \rangle$	$\sqrt{1 + g_x^2 + g_y^2}$

Common Mistakes:

- Wrong orientation: check that your normal points in the required direction before integrating
- Forgetting to include $|\mathbf{r}_u \times \mathbf{r}_v|$ in scalar surface integrals
- On $z = g(x, y)$: the downward normal is $\langle g_x, g_y, -1 \rangle$ (signs flip)

4.6 Vector Calculus

Line Integrals

Scalar: $\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$

Vector (work): $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$

Conservative Fields and Potential Functions

$\mathbf{F} = \langle P, Q, R \rangle$ is conservative on a simply connected domain iff $\nabla \times \mathbf{F} = \mathbf{0}$ (i.e., $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, etc.)

Fundamental Theorem for Line Integrals:

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Finding potential f : Integrate P w.r.t. x , then differentiate w.r.t. y and match Q .

Curl and Divergence

$$\nabla \times \mathbf{F} = \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}$$

$$\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

The Four Big Theorems

Green's Theorem (2D, relates line integral around closed curve C to double integral over enclosed region D):

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Stokes' Theorem (generalizes Green's to 3D surface S with boundary C ; for setting up $d\mathbf{S}$, see §4.5b):

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

Divergence Theorem (relates flux through closed surface S to volume integral over enclosed E):

$$\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E (\nabla \cdot \mathbf{F}) dV$$

When to use each:

- Green's: 2D closed curve, hard to integrate directly
- Stokes': 3D closed curve, or simplify a surface integral to a different surface
- Divergence: closed surface, outward flux, often easier than direct surface integration

Common Mistakes:

- Spherical: mixing up ϕ (from z -axis) and θ (azimuthal), and forgetting $\sin \phi$ in dV
- Green's: wrong sign ($Q_x - P_y$ not $P_x - Q_y$)
- Divergence theorem: applying to non-closed surfaces

SUBJECT 5: DIFFERENTIAL EQUATIONS

5.1 First-Order ODEs

Separable Equations

Form: $\frac{dy}{dx} = g(x)h(y)$

Procedure: $\frac{dy}{h(y)} = g(x) dx$, then integrate both sides.

Example: $\frac{dy}{dx} = xy, y(0) = 2$

$$\frac{dy}{y} = x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C_1 \Rightarrow y = Ce^{x^2/2}$$

$$y(0) = 2 \Rightarrow C = 2 \Rightarrow y = 2e^{x^2/2}$$

Linear First-Order: $\frac{dy}{dx} + P(x)y = Q(x)$

Integrating factor: $\mu(x) = e^{\int P(x) dx}$

General solution: $y = \frac{1}{\mu(x)} \int \mu(x)Q(x) dx + \frac{C}{\mu(x)}$

Equivalently: multiply both sides by μ ; left side becomes $\frac{d}{dx}[\mu y]$; integrate.

Example: $y' + \frac{2}{x}y = 4x, y(1) = 2$

$$\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$$

$$\frac{d}{dx}[x^2 y] = 4x \cdot x^2 = 4x^3$$

$$x^2 y = x^4 + C \Rightarrow y = x^2 + Cx^{-2}$$

$$y(1) = 1 + C = 2 \Rightarrow C = 1 \Rightarrow y = x^2 + x^{-2}$$

Exact Equations

Form: $M(x, y) dx + N(x, y) dy = 0$

Test for exactness: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Procedure: Find $F(x, y)$ such that $F_x = M$ and $F_y = N$:

1. $F = \int M dx + g(y)$
2. Differentiate w.r.t. y : $F_y = (\int M dx)_y + g'(y) = N$
3. Solve for $g'(y)$, integrate for $g(y)$
4. Solution: $F(x, y) = C$

Example: $(2xy - y^2) dx + (x^2 - 2xy) dy = 0$

$$M_y = 2x - 2y = N_x \checkmark$$

$$F = \int (2xy - y^2) dx = x^2y - xy^2 + g(y)$$

$$F_y = x^2 - 2xy + g'(y) = x^2 - 2xy \Rightarrow g'(y) = 0 \Rightarrow g = C_1$$

$$\text{Solution: } x^2y - xy^2 = C$$

Bernoulli Equations

Form: $y' + P(x)y = Q(x)y^n$ (nonlinear when $n \neq 0, 1$)

Substitution: $v = y^{1-n}$ transforms to linear ODE:

$$v' + (1-n)P(x)v = (1-n)Q(x)$$

Homogeneous Equations (Type: $dy/dx = f(y/x)$)

Substitution: $v = y/x$ (so $y = vx$, $y' = v + xv'$) $\rightarrow$ separable in v and x .

Existence and Uniqueness

For $y' = f(x, y)$, $y(x_0) = y_0$: if f and $\partial f / \partial y$ are continuous near (x_0, y_0) , a unique solution exists in some interval around x_0 .

Common Mistakes:

- Linear ODE: computing μ incorrectly from $P(x)$ (make sure equation is in standard form first)
- Exact: computing $g'(y)$ and finding x terms remain (not exact, or computational error)

5.2 Second-Order Linear ODEs with Constant Coefficients

Homogeneous: $ay'' + by' + cy = 0$

Characteristic equation: $ar^2 + br + c = 0$, $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\Delta = b^2 - 4ac$	Roots	General solution y_h
$\Delta > 0$	Two distinct real r_1, r_2	$C_1 e^{r_1 x} + C_2 e^{r_2 x}$
$\Delta = 0$	Repeated real root r	$(C_1 + C_2 x) e^{rx}$

$\Delta = b^2 - 4ac$	Roots	General solution y_h
$\Delta < 0$	Complex $\alpha \pm \beta i$	$e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

Example: $y'' - 5y' + 6y = 0$

$$r^2 - 5r + 6 = 0 \Rightarrow (r - 2)(r - 3) = 0 \Rightarrow r = 2, 3$$

$$y = C_1 e^{2x} + C_2 e^{3x}$$

Example (complex): $y'' + 4y = 0$

$$r^2 + 4 = 0 \Rightarrow r = \pm 2i \ (\alpha = 0, \beta = 2)$$

$$y = C_1 \cos 2x + C_2 \sin 2x$$

Non-Homogeneous: $ay'' + by' + cy = g(x)$

General solution: $y = y_h + y_p$

Method of Undetermined Coefficients (guess table for y_p):

$g(x)$ form	Guess for y_p
$P_n(x)$ (degree n poly)	$x^s(A_n x^n + \dots + A_0)$
$e^{\alpha x}$	$x^s A e^{\alpha x}$
$\cos(\beta x)$ or $\sin(\beta x)$	$x^s(A \cos \beta x + B \sin \beta x)$
$e^{\alpha x} \cos(\beta x)$ or $e^{\alpha x} \sin(\beta x)$	$x^s e^{\alpha x}(A \cos \beta x + B \sin \beta x)$
$P_n(x)e^{\alpha x}$	$x^s e^{\alpha x}(A_n x^n + \dots + A_0)$

Modification rule: s = smallest non-negative integer such that no term in y_p duplicates y_h . Usually $s = 0$; use $s = 1$ if the exponential matches a root, $s = 2$ if it's a repeated root.

Example: $y'' - 3y' + 2y = 4e^{2x}$

$$y_h: r^2 - 3r + 2 = 0, (r - 1)(r - 2) = 0, y_h = C_1 e^x + C_2 e^{2x}$$

Since e^{2x} appears in y_h : guess $y_p = A x e^{2x}$ (use $s = 1$)

$$y_p' = A e^{2x}(1 + 2x), y_p'' = A e^{2x}(4 + 4x)$$

$$\text{Substitute: } A e^{2x}(4 + 4x) - 3A e^{2x}(1 + 2x) + 2A x e^{2x} = 4e^{2x}$$

$$A e^{2x}(4 + 4x - 3 - 6x + 2x) = A e^{2x} = 4e^{2x} \Rightarrow A = 4$$

$$y_p = 4x e^{2x}$$

Variation of Parameters (general — works for any $g(x)$)

Given $y_h = C_1 y_1 + C_2 y_2$:

$$\text{Wronskian: } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$u_1' = \frac{-y_2 g}{aW}, \quad u_2' = \frac{y_1 g}{aW}$$

Integrate to get u_1, u_2 . Then $y_p = u_1 y_1 + u_2 y_2$.

Reduction of Order

If y_1 is a known solution, let $y_2 = v(x)y_1$, substitute into the ODE, and solve the resulting first-order ODE in v' .

Common Mistakes:

- UDC: not checking if y_p guess duplicates y_h before differentiating
- Variation of parameters: using $g(x)$ directly without putting the ODE in standard form first (divide by a so coefficient of y'' is 1)

5.3 Laplace Transforms

Definition

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Complete Transform Table

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Notes
1	$1/s$	$s > 0$
t^n	$n!/s^{n+1}$	$s > 0$
e^{at}	$1/(s-a)$	$s > a$
$t^n e^{at}$	$n!/(s-a)^{n+1}$	
$\sin(bt)$	$b/(s^2 + b^2)$	
$\cos(bt)$	$s/(s^2 + b^2)$	
$e^{at} \sin(bt)$	$b/((s-a)^2 + b^2)$	
$e^{at} \cos(bt)$	$(s-a)/((s-a)^2 + b^2)$	
$\sinh(bt)$	$b/(s^2 - b^2)$	
$\cosh(bt)$	$s/(s^2 - b^2)$	
$u(t-a)$	e^{-as}/s	unit step
$\delta(t-a)$	e^{-as}	Dirac delta
$f(t-a)u(t-a)$	$e^{-as}F(s)$	2nd shift theorem
$e^{at}f(t)$	$F(s-a)$	1st shift theorem
$f'(t)$	$sF(s) - f(0)$	
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
$f^{(n)}(t)$	$s^n F(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$	

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Notes
$(f * g)(t) = \int_0^t f(\tau)g(t - \tau)d\tau$	$F(s)G(s)$	convolution

Solving IVPs with Laplace

Procedure:

1. Take $\mathcal{L}$ of both sides
2. Apply derivative property to express $\mathcal{L}\{y'\}$ and $\mathcal{L}\{y''\}$ in terms of $Y(s) = \mathcal{L}\{y\}$ and ICs
3. Solve algebraically for $Y(s)$
4. Apply partial fractions and inverse Laplace to find $y(t)$

Example: $y'' + 3y' + 2y = e^{-t}$, $y(0) = 0$, $y'(0) = 1$

Take Laplace: $[s^2Y - s \cdot 0 - 1] + 3[sY - 0] + 2Y = \frac{1}{s+1}$

$$(s^2 + 3s + 2)Y = 1 + \frac{1}{s+1} = \frac{s+2}{s+1}$$

$$(s+1)(s+2)Y = \frac{s+2}{s+1}, \text{ so } Y = \frac{1}{(s+1)^2}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = te^{-t}$$

Inverse Laplace — Key Techniques

1. **Partial fractions:** decompose $Y(s)$ into recognizable forms from the table
2. **Complete the square:** for complex poles, get $(s - a)^2 + b^2$ in denominator
3. **First shift theorem:** $\mathcal{L}^{-1}\{F(s - a)\} = e^{at}f(t)$

Unit Step and Discontinuous Forcing

$$u(t - a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

$$\mathcal{L}\{u(t - a)g(t)\} = e^{-as}\mathcal{L}\{g(t + a)\}$$

Common Mistakes:

- Applying the 1st shift theorem backwards: $F(s - a)$ in s -domain corresponds to e^{at} (positive exponent) in time domain
- Forgetting initial condition terms in $\mathcal{L}\{y''\}$

5.4 Systems of First-Order Linear ODEs

Matrix Form

Convert to $\mathbf{x}' = A\mathbf{x}$ where $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Converting a 2nd-order ODE: For $y'' + py' + qy = 0$, let $x_1 = y$, $x_2 = y'$:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Solution via Eigenvalues

Eigenvalue/eigenvector procedure: see §6.5. The three cases below (distinct real, repeated, complex) mirror the three cases for 2nd-order ODEs in §5.2.

Find eigenvalues from $\det(A - \lambda I) = 0$.

Case 1 — Distinct real eigenvalues λ_1, λ_2 with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{x} = C_1 \mathbf{v}_1 e^{\lambda_1 t} + C_2 \mathbf{v}_2 e^{\lambda_2 t}$$

Case 2 — Repeated eigenvalue λ (algebraic multiplicity 2):

If only one eigenvector $\mathbf{v}_1$ exists, find generalized eigenvector $\mathbf{v}_2$: $(A - \lambda I)\mathbf{v}_2 = \mathbf{v}_1$

$$\mathbf{x} = C_1 \mathbf{v}_1 e^{\lambda t} + C_2 (\mathbf{v}_1 t + \mathbf{v}_2) e^{\lambda t}$$

Case 3 — Complex eigenvalues $\alpha \pm \beta i$ with eigenvector $\mathbf{v} = \mathbf{a} + i\mathbf{b}$:

$$\mathbf{x} = C_1 e^{\alpha t} (\mathbf{a} \cos \beta t - \mathbf{b} \sin \beta t) + C_2 e^{\alpha t} (\mathbf{a} \sin \beta t + \mathbf{b} \cos \beta t)$$

Example: $\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{x}$

$$\det(A - \lambda I) = (1 - \lambda)(2 - \lambda) - 6 = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1) = 0$$

$$\lambda_1 = 4: (A - 4I)\mathbf{v} = 0 \Rightarrow \mathbf{v}_1 = \langle 2, 3 \rangle^T$$

$$\lambda_2 = -1: (A + I)\mathbf{v} = 0 \Rightarrow \mathbf{v}_2 = \langle -1, 1 \rangle^T$$

$$\mathbf{x} = C_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{4t} + C_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}$$

5.5 Numerical Methods for ODEs

Euler's Method

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

h = step size. Global error $\sim O(h)$.

Improved Euler / Heun's Method

$$k_1 = f(x_n, y_n), \quad k_2 = f(x_n + h, y_n + hk_1)$$

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2)$$

Global error $\sim O(h^2)$.

Runge-Kutta 4th Order (RK4)

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + h/2, y_n + hk_1/2)$$

$$k_3 = f(x_n + h/2, y_n + hk_2/2)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Global error $\sim O(h^4)$. Standard choice for most problems.

5.6 Qualitative Analysis

Autonomous Equations: $\frac{dy}{dt} = f(y)$

Equilibrium solutions: values y^* where $f(y^*) = 0$

Stability:

- **Stable (attractor):** $f'(y^*) < 0$ — nearby solutions converge
- **Unstable (repeller):** $f'(y^*) > 0$ — nearby solutions diverge
- **Semi-stable:** f doesn't change sign at y^*

Phase line: plot $f(y)$ vs. y ; where $f > 0$, solutions increase; where $f < 0$, solutions decrease.

Example: $dy/dt = y(1-y)(2-y)$

Equilibria: $y = 0, 1, 2$. Check signs of f in each interval:

Interval	y	$1-y$	$2-y$	f	Direction
$y < 0$	—	+	+	—	decreasing
$0 < y < 1$	+	+	+	+	increasing
$1 < y < 2$	+	—	+	—	decreasing
$y > 2$	+	—	—	+	increasing

Result: $y = 0$ unstable, $y = 1$ stable, $y = 2$ unstable.

SUBJECT 6: LINEAR ALGEBRA

6.1 Vectors and Vector Spaces

Vector Space Axioms

A set V with operations $+$ and scalar multiplication satisfies all 10 axioms (closure, commutativity, associativity, identity, inverse, distributivity, etc.). Key examples: $\mathbb{R}^n$, P_n (polynomials of degree $\leq n$), $M_{m \times n}$ (matrices).

Subspace Test

$H \subseteq V$ is a subspace iff:

1. $\mathbf{0} \in H$
2. Closed under addition: $\mathbf{u}, \mathbf{v} \in H \Rightarrow \mathbf{u} + \mathbf{v} \in H$
3. Closed under scalar multiplication: $\mathbf{u} \in H, c \in \mathbb{R} \Rightarrow c\mathbf{u} \in H$

Linear Independence

$\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent iff $c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p = \mathbf{0}$ implies $c_1 = \dots = c_p = 0$.

Test: set up the matrix $[\mathbf{v}_1 \dots \mathbf{v}_p]$ and row reduce. Independent iff no free variables.

Basis and Dimension

A **basis** for H is a linearly independent set that spans H . The **dimension** is the number of vectors in any basis.

Rank-Nullity Theorem

For $m \times n$ matrix A :

$$\text{rank}(A) + \text{nullity}(A) = n$$

where $\text{rank} = \dim(\text{col space}) = \text{number of pivot columns}$, $\text{nullity} = \dim(\text{null space}) = \text{number of free variables}$.

6.2 Matrices and Operations

Matrix Multiplication

$$(AB)_{ij} = \sum_k A_{ik} B_{kj} \text{ (row of } A \text{ dotted with column of } B\text{)}$$

AB is defined only if columns of A = rows of B . $AB \neq BA$ in general.

Properties:

- $(AB)C = A(BC)$ (associative)
- $A(B + C) = AB + AC$ (distributive)
- $(AB)^T = B^T A^T$
- $(AB)^{-1} = B^{-1} A^{-1}$

Matrix Inverse

A^{-1} exists iff $\det(A) \neq 0$ iff A has n pivot positions.

$$\mathbf{2 \times 2 \text{ formula:}} \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

General method: Row reduce $[A|I] \rightarrow [I|A^{-1}]$

6.3 Systems of Linear Equations

Gaussian Elimination to REF

Row operations (preserve solution set):

1. Swap two rows
2. Multiply a row by a nonzero scalar
3. Add a multiple of one row to another

REF: each leading entry is to the right of all leading entries above it.

RREF: each leading entry is 1, and it's the only nonzero entry in its column.

Solution structure: number of solutions determined by consistency and free variables.

- Inconsistent (row $[0 \ \cdots \ 0 \mid b]$, $b \neq 0$): no solution
- Consistent, no free variables: exactly one solution
- Consistent, k free variables: infinitely many solutions (parametric form, k -dimensional solution set)

Example: Solve
$$\begin{cases} x + 2y + z = 4 \\ 2x + 5y + 3z = 9 \\ x + 3y + 2z = 5 \end{cases}$$

Augmented matrix row reduction:

$$\left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 2 & 5 & 3 & 9 \\ 1 & 3 & 2 & 5 \end{array}\right) \xrightarrow{R_2-2R_1, R_3-R_1} \left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow{R_3-R_2} \left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Free variable: $z = t$. Back-substitute: $y = 1 - t$, $x = 4 - 2(1 - t) - t = 2 + t$.

Solution: $(x, y, z) = (2, 1, 0) + t(1, -1, 1)$.

6.4 Determinants

Cofactor Expansion (along row i)

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is the minor (det of the $(n-1) \times (n-1)$ submatrix with row i and column j removed).

Sign pattern:
$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Example: $\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, expand along row 1:

$$= 1 \det \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix} - 2 \det \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix} + 3 \det \begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix}$$

$$= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0 \text{ (rows are linearly dependent)}$$

Properties

- $\det(AB) = \det(A) \det(B)$
- $\det(A^T) = \det(A)$
- $\det(A^{-1}) = 1/\det(A)$
- Swapping two rows multiplies det by -1
- Multiplying a row by k multiplies det by k
- Adding a multiple of one row to another: det unchanged
- $\det(A) = 0 \Leftrightarrow A$ is singular (not invertible)

Cramer's Rule

For $n \times n$ system $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is A with column i replaced by $\mathbf{b}$.

Common Mistakes:

- Getting the sign pattern wrong in cofactor expansion
- $\det(kA) = k^n \det(A)$ (not $k \det(A)$)

6.5 Eigenvalues and Eigenvectors

Definition and Procedure

λ is an eigenvalue of A and $\mathbf{v} \neq \mathbf{0}$ is the corresponding eigenvector iff $A\mathbf{v} = \lambda\mathbf{v}$, equivalently $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

Step 1: Solve $\det(A - \lambda I) = 0$ (characteristic equation) for λ .

Step 2: For each λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ by row reduction. Each free variable gives one eigenvector.

Example: $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$

$$\det(A - \lambda I) = (3 - \lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2)$$

$$\lambda_1 = 4: (A - 4I) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 2: (A - 2I) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Eigenvalues drive the solution to systems of ODEs — see §5.4 for the direct connection.

Algebraic vs. Geometric Multiplicity

- **Algebraic multiplicity:** multiplicity of λ as a root of the characteristic polynomial
- **Geometric multiplicity:** $\dim(\text{null}(A - \lambda I))$ = number of linearly independent eigenvectors for λ
- Always: geometric $\leq$ algebraic

Diagonalization

$A = PDP^{-1}$ where D is diagonal (eigenvalues on diagonal) and P has eigenvectors as columns.

Condition: A is diagonalizable iff it has n linearly independent eigenvectors (equivalently, geometric multiplicity = algebraic multiplicity for each eigenvalue).

Powers: $A^k = PD^kP^{-1}$ where D^k has λ_i^k on the diagonal.

Symmetric matrices: always diagonalizable; eigenvectors for distinct eigenvalues are orthogonal; can diagonalize as $A = QDQ^T$ where Q is orthogonal ($Q^{-1} = Q^T$).

Common Mistakes:

- Putting eigenvectors in the wrong order in P — they must match the order of eigenvalues in D
- A repeated eigenvalue does not automatically mean the matrix is not diagonalizable — check geometric multiplicity

6.6 Orthogonality

Orthogonal and Orthonormal Sets

$\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is orthogonal if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$.

Orthonormal: additionally $|\mathbf{u}_i| = 1$ for all i .

An orthogonal set of nonzero vectors is automatically linearly independent.

Gram-Schmidt Process

Given basis $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, produce orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$:

$$\begin{aligned}\mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{|\mathbf{v}_1|^2} \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{|\mathbf{v}_1|^2} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{|\mathbf{v}_2|^2} \mathbf{v}_2\end{aligned}$$

Then normalize: $\mathbf{u}_i = \mathbf{v}_i / |\mathbf{v}_i|$

QR Decomposition

$A = QR$ where Q has orthonormal columns (from Gram-Schmidt on columns of A) and R is upper triangular.

$$R = Q^T A \text{ (since } Q^T Q = I\text{)}$$

Orthogonal Projection

Projection of $\mathbf{y}$ onto subspace W with orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$:

$$\hat{\mathbf{y}} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

Projection matrix: $P = QQ^T$ where $Q = [\mathbf{u}_1 \cdots \mathbf{u}_p]$. Then $\hat{\mathbf{y}} = P\mathbf{y}$.

Least Squares

For overdetermined $A\mathbf{x} = \mathbf{b}$ (no solution), minimize $|\mathbf{b} - A\mathbf{x}|$:

Normal equations: $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$

If A has linearly independent columns: $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$

Example: Fit a line $y = \beta_0 + \beta_1 x$ to points $(0, 1), (1, 3), (2, 4), (3, 5)$.

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, A^T \mathbf{b} = \begin{pmatrix} 13 \\ 26 \end{pmatrix}$$

Solve: $\hat{\beta}_0 = 1.5, \hat{\beta}_1 = 4/3 \approx 1.33$

6.7 Linear Transformations

Definition

$T : V \rightarrow W$ is linear iff:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$

Equivalently: $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ and $T(\mathbf{0}) = \mathbf{0}$.

Matrix Representation

Every linear $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has $T(\mathbf{x}) = A\mathbf{x}$ where $A = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$ ($m \times n$).

Standard transformations in $\mathbb{R}^2$:

- Rotation by θ : $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
- Reflection over x -axis: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- Projection onto x -axis: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Kernel and Image

$$\ker(T) = \{\mathbf{x} : T(\mathbf{x}) = \mathbf{0}\} = \text{null}(A)$$

$$\text{im}(T) = \{T(\mathbf{x}) : \mathbf{x} \in V\} = \text{col}(A)$$

- T is **one-to-one** (injective) iff $\ker(T) = \{\mathbf{0}\}$
- T is **onto** (surjective) iff $\text{im}(T) = W$

Change of Basis

If $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $B' = \{\mathbf{b}'_1, \dots, \mathbf{b}'_n\}$ are bases, the change-of-basis matrix from B to the standard basis is $P_B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$.

$[T]_{B'} = P^{-1}[T]P$ where P is the change-of-basis matrix.

6.8 Matrix Decompositions

LU Decomposition

$A = LU$ where L is lower triangular (1's on diagonal) and U is upper triangular.

Obtained by tracking the row operations in Gaussian elimination. Used to solve $A\mathbf{x} = \mathbf{b}$ efficiently: forward solve $L\mathbf{y} = \mathbf{b}$, then back-substitute $U\mathbf{x} = \mathbf{y}$.

Singular Value Decomposition (SVD)

For any $m \times n$ matrix A :

$$A = U\Sigma V^T$$

- U : $m \times m$ orthogonal matrix (columns = left singular vectors — orthonormal basis for $\mathbb{R}^m$)
- Σ : $m \times n$ diagonal matrix with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ on diagonal
- V : $n \times n$ orthogonal matrix (columns = right singular vectors — orthonormal basis for $\mathbb{R}^n$)

Computing SVD:

- Singular values $\sigma_i = \sqrt{\lambda_i(A^T A)}$
- V = eigenvectors of $A^T A$
- U = eigenvectors of AA^T

Applications:

- Rank of A = number of nonzero singular values
- Pseudoinverse: $A^+ = V\Sigma^+U^T$ where Σ^+ replaces each nonzero σ_i with $1/\sigma_i$
- Least squares: $\hat{\mathbf{x}} = A^+\mathbf{b}$
- Data compression / low-rank approximation: retain top k singular values

Common Mistakes:

- Gram-Schmidt: not re-orthogonalizing against ALL previous vectors in each step
- SVD: mixing up left and right singular vectors (U vs. V)
- Least squares: applying normal equations when $A^T A$ is singular (use SVD instead)